INCH-POUND

MIL-DTL-16878/36A
11 August 2000
SUPERSEDING
MIL-W-16878/36(NAVY)
11 September 1992

DETAIL SPECIFICATION SHEET

WIRE, ELECTRICAL, LOW SMOKE, CROSSLINKED POLYOLEFIN (XLPO) INSULATED, 105 °C, 600 VOLTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16878G.

REQUIREMENTS.

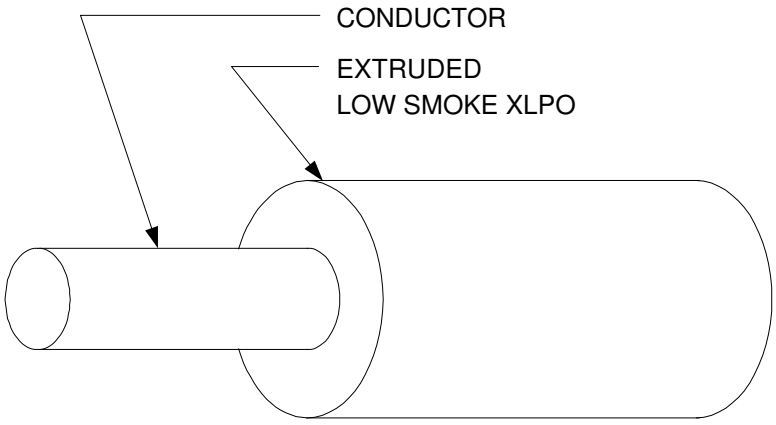

FIGURE 1. Wire configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Wire construction details.

PIN [1/]	Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material [2/, 3/]	Coating		Min	Max
M16878/36BAA*	32	1 X 32	Copper	Tin	.0080	.024	.030
M16878/36CAA*	32	1 X 32	C.C. steel	Tin	.0080	.024	.030
M16878/36BAB*	32	7 X 40	Copper	Tin	.0100	.026	.032
M16878/36DAB*	32	7 X 40	H.S.C.A	Silver	.0100	.026	.032
M16878/36BBA*	30	1 X 30	Copper	Tin	.0100	.026	.032
M16878/36DBA*	30	1 X 30	H.S.C.A	Silver	.0100	.026	.032
M16878/36CBA*	30	1 X 30	C.C. steel	Tin	.0100	.026	.032
M16878/36BBB*	30	7 X 38	Copper	Tin	.0120	.026	.034
M16878/36DBB*	30	7 X 38	H.S.C.A	Silver	.0120	.028	.034
M16878/36BCA*	28	1 X 28	Copper	Tin	.0126	.028	.035
M16878/36DCA*	28	1 X 28	H.S.C.A	Silver	.0126	.029	.035
M16878/36CCA*	28	1 X 28	C.C. steel	Tin	.0126	.029	.035
M16878/36BCB*	28	7 X 36	Copper	Tin	.0150	.029	.037
M16878/36DCB*	28	7 X 36	H.S.C.A	Silver	.0150	.031	.037
M16878/36BDA*	26	1 X 26	Copper	Tin	.0159	.031	.038
M16878/36DDA*	26	1 X 26	H.S.C.A	Silver	.0159	.032	.038
M16878/36CDA*	26	1 X 26	C.C. steel	Tin	.0159	.032	.038
M16878/36BDB*	26	7 X 34	Copper	Tin	.0190	.032	.041
M16878/36DDB*	26	7 X 34	H.S.C.A	Silver	.0190	.035	.041
M16878/36BDE*	26	19 X 38	Copper	Tin	.0210	.035	.041
M16878/36DDE*	26	19 X 38	H.S.C.A	Silver	.0210	.035	.041
M16878/36BEA*	24	1 X 24	Copper	Tin	.0201	.035	.043
M16878/36DEA*	24	1 X 24	H.S.C.A	Silver	.0201	.036	.043
M16878/36CEA*	24	1 X 24	C.C. steel	Tin	.0201	.036	.043
M16878/36BEB*	24	7 X 32	Copper	Tin	.0240	.036	.047
M16878/36DEB*	24	7 X 32	H.S.C.A	Silver	.0240	.040	.047
M16878/36BEE*	24	19 X 36	Copper	Tin	.0260	.040	.047
M16878/36DEE*	24	19 X 36	H.S.C.A	Silver	.0260	.040	.047
M16878/36BFA*	22	1 X 22	Copper	Tin	.0253	.041	.048
M16878/36DFA*	22	1 X 22	H.S.C.A	Silver	.0253	.041	.048
M16878/36CFA*	22	1 X 22	C.C. steel	Tin	.0253	.041	.048
M16878/36BFB*	22	7 X 30	Copper	Tin	.0300	.046	.053
M16878/36DFB*	22	7 X 30	H.S.C.A	Silver	.0300	.046	.053
M16878/36BFE*	22	19 X 34	Copper	Tin	.0320	.046	.053
M16878/36DFE*	22	19 X 34	H.S.C.A	Silver	.0320	.046	.053
M16878/36BGA*	20	1 X 20	Copper	Tin	.0320	.048	.055

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TABLE I. Wire configuration and dimensions - Continued.

PIN 1/	Wire size	Stranding	Conductor		Conductor Diameter (nominal) (inch)	Finished wire diameter (inch)	
			Material 2/, 3/	Coating		Min	Max
M16878/36DGA*	20	1 X 20	H.S.C.A	Silver	.0320	.048	.055
M16878/36BGB*	20	7 X 28	Copper	Tin	.0380	.054	.061
M16878/36DGB*	20	7 X 28	H.S.C.A	Silver	.0380	.054	.061
M16878/36BGE*	20	19 X 32	Copper	Tin	.0410	.054	.061
M16878/36BHA*	18	1 X 18	Copper	Tin	.0403	.056	.063
M16878/36BHB*	18	7 X 26	Copper	Tin	.0490	.065	.072
M16878/36BHE*	18	19 X 30	Copper	Tin	.0510	.065	.072
M16878/36BJA*	16	1 X 16	Copper	Tin	.0508	.067	.075
M16878/36BJE*	16	19 X 29	Copper	Tin	.0590	.075	.083
M16878/36BKA*	14	1 X 14	Copper	Tin	.0641	.080	.088
M16878/36BKE*	14	19 X 27	Copper	Tin	.0720	.088	.096

Notes:
1/ PIN stands for part or identifying number (see figure 2).
2/ H.S.C.A. stands for high-strength copper alloy.
3/ C.C stands for copper-clad.

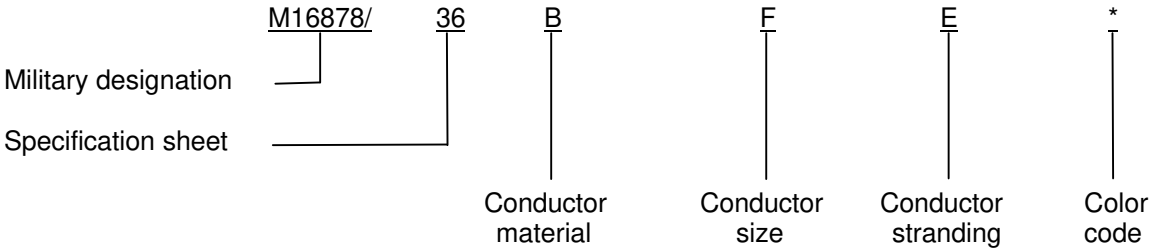

FIGURE 2. Example of PIN (see MIL-DTL-16878G).

Configuration and dimensions: See figure 1 and table I
Operating voltage: Up to 600 volts
Operating temperature: Up to 105 °C
Insulation: Low smoke crosslinked polyolefin
Spark test voltage: 3.4 kV
Impulse dielectric test voltage: 8.0 kV, or 5.7 kV using the 3.0 kHz spark test
Dielectric withstanding voltage: 2.0 kV
Insulation resistance: $IR = K \log_{10} D/d$

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20 °C
K = 2,000
D = Maximum average diameter of finished wire
d = Conductor diameter

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Cold bend: Condition 4 hours at -40±1 °C (see table II)

TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel diameter (inches, maximum)
32 through 26	1
24 through 14	2

Surface resistance: Not required
Heat resistance: Condition at 150 °C
Heat aging: 25 percent change (maximum) in 96 hours at 135 °C
Insulation tensile strength: 1800 pounds force per square inch (minimum)
Insulation elongation: 100 percent (minimum)
Marking and stripe durability: Required when tested in accordance with MIL-C-24643
Smoke index: 45 (maximum) when tested in accordance with MIL-C-24643
Toxicity index: 1.5 (maximum) when tested in accordance with MIL-C-24643
Acid gas equivalent: 18 percent (maximum) when tested in accordance with MIL-C-24643

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue because of the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Navy - SH
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2193-018)

Review activity:
Navy - AS

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